Unpacking Fairness in Recommendation

Statistical Guarantees, User Control, and Collective Resistance

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RecSoGood @ RecSys '26 - Minneapolis, MN, US, October 2, 2026

European Commission

Link to SDGs


10
REDUCED
INEQUALITIES

12
RESPONSIBLE
CONSUMPTION
AND PRODUCTION

16
PEACE, JUSTICE
AND STRONG
INSTITUTIONS

Link to SDGs



Mitigating algorithmic bias and disparate impact across user groups.

Link to SDGs



Empowering users to curate healthy,
non-distressing digital consumption choices.

Link to SDGs





Auditing online safety mechanisms against malicious coordination and systemic manipulation.

Sources

You Don't Bring Me Flowers: Mitigating Unwanted Recommendations Through Conformal Risk Control

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RecSys 2025 Prague

Best Full Paper Award

[1] De Toni, G., Purificato, E., Gomez, E., Passerini, A., Lepri, B. and Consonni, C., 2025, September. You Don't Bring Me Flowers: Mitigating Unwanted Recommendations Through Conformal Risk Control. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems* (pp. 492-502).

Sources

With a Little Help From My Friends: Collective Manipulation in Risk-Controlling Recommender Systems

GIOVANNI DE TONI, Fondazione Bruno Kessler (FBK), Italy
CRISTIAN CONSONNI*, European Commission, Joint Research Centre (JRC), Italy
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[2] De Toni, G., Consonni, C., Purificato, E., Gómez, E. and Lepri, B., 2026, June. With a Little Help From My Friends: Collective Manipulation in Risk-Controlling Recommender Systems. In *The 2026 ACM Conference on Fairness, Accountability, and Transparency* (pp. 4712-4732).

Chapter 1.

The problem

~5 hours

Daily teen social media usage [3].

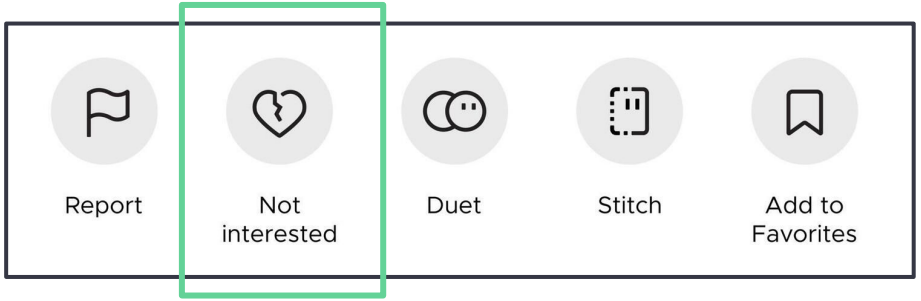
Negative impact on mental health and performance [4,5].

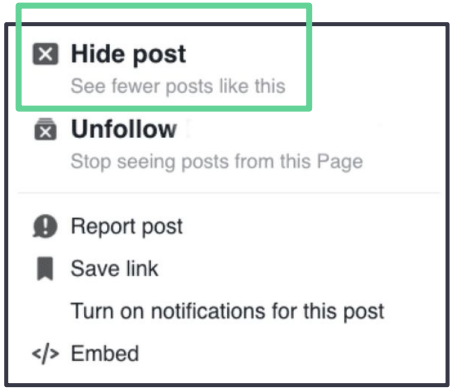
Unwanted content

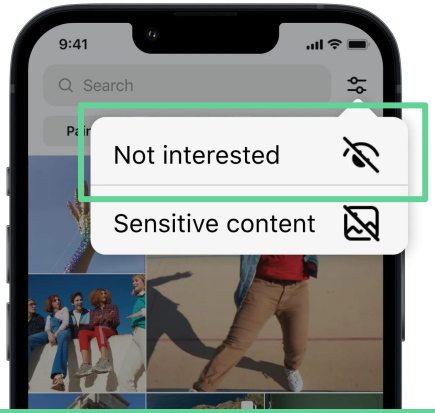
Recommenders contribute to the spread of it [6].

Risk of generating filter bubbles or rabbit holes [7].


Save
Copy link to post
Embed this post
Unfollow
Not interested


Report
Not interested
Duet
Stitch
Add to Favorites


Hide post
See fewer posts like this
Unfollow
Stop seeing posts from this Page
Report post
Save link
Turn on notifications for this post
Embed


9:41
Search
Not interested
Sensitive content


Save
Copy link to post
Embed this p
Unfollow
Not interested
Report
Add to
Favorites
Hide post
See fewer posts like this
Unfollow
seeing posts from this Page
post
t interested
Sensitive content

Illusion of control

Low user awareness or limited effectiveness [8].

Lack of theoretical guarantees and practical methods [9].

Spread of folk theories about influencing a recommender behavior [10].

Watch time (alone [11])

is **NOT** an indicator of preferences.



The Guardian, September 3, 2025.
https://www.theguardian.com/lifeandstyle/2025/sep/03/i-felt-doomed-social-media-guessed-i-was-pregnant-my-feed-soon-grew-horrifying.

Rubbernecking

Slowing down or turning your head to stare at an accident, emergency vehicle, or unusual roadside event.

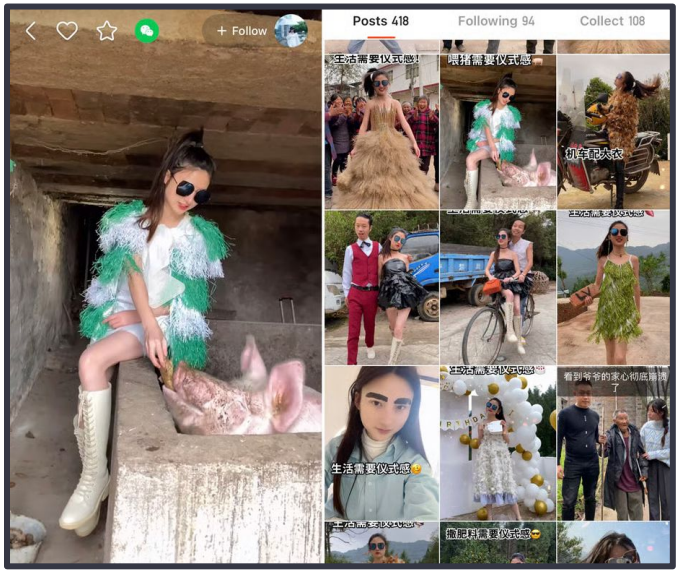
KuaiRand dataset

- From **Kuaishou** (https://www.kuaishou.com/)

  300M daily users in 2025!
- 27,000 users
- 32M unique videos watched

  (April 8 to May 8, 2022)
- ***Do Not Recommend*** signal!

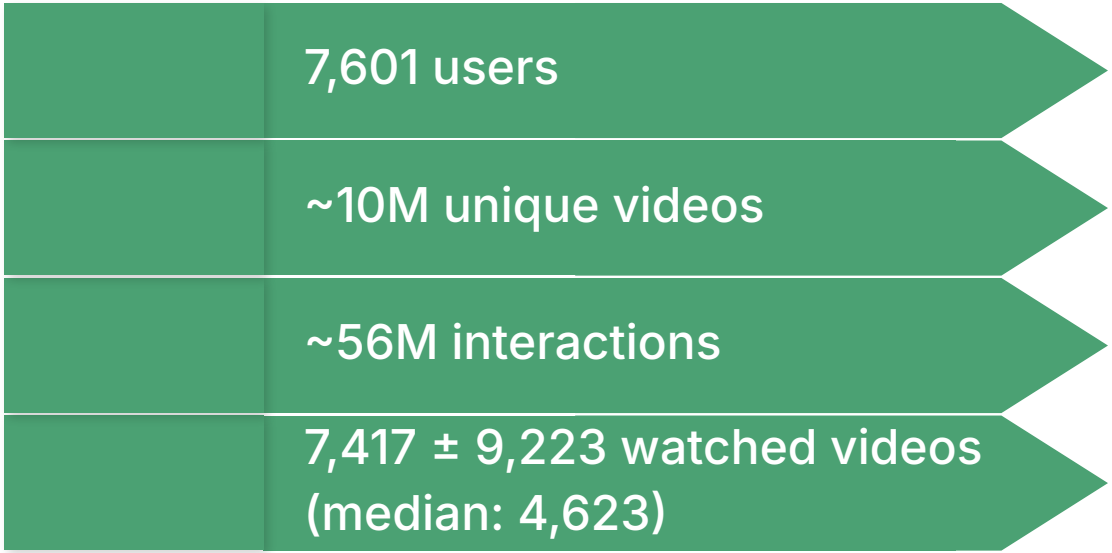
Kuaishou



[12] Gao, C., Li, S., Zhang, Y., Chen, J., Li, B., Lei, W., Jiang, P. and He, X., 2022, October. Kuairand: An unbiased sequential recommendation dataset with randomly exposed videos. In Proceedings of the 31st ACM international conference on information & knowledge management (pp. 3953-3957).

Our filtered data source

- Removed **ads** and **zero-duration** videos
- Users w/ **negative feedback** (29%)



Trends of video-reporting patterns

Behaviour	I (%)	U (%)	Watch Time 1st (s)	Watch Time 2nd (s)
✓ → ✓	99.79	100.0	2.71 ±12.15 (1.39)	10.48 ±25.85 (7.95)
✓ → ✗	0.11	8.69	0.50 ±7.41 (0.0)	10.73 ±20.85 (10.13)
✗ → ✗	0.07	6.33	7.74 ±15.83 (6.64)	7.85 ±15.86 (6.72)
✗ → ✓	0.03	1.49	1.36 ±6.91 (0.0)	6.46 ±25.42 (2.5)

Users usually **do not** report unwanted/harmful content.

Trends of video-reporting patterns

Behaviour	I (%)	U (%)	Watch Time 1st (s)	Watch Time 2nd (s)
✓ → ✓	99.79	100.0	2.71 ±12.15 (1.39)	10.48 ±25.85 (7.95)
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Platforms can recommend unwanted content **despite explicit feedback**.

Trends of video-reporting patterns

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✓ → ✓	99.79	100.0	2.71 ±12.15 (1.39)	10.48 ±25.85 (7.95)
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Very short consumption can only be an **early indicator** of undesidered content.

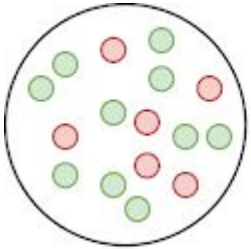
How can we (provably) **mitigate exposure** to **unwanted recommendations** given (binary) user's feedback?

Chapter 2.

Our initial solution

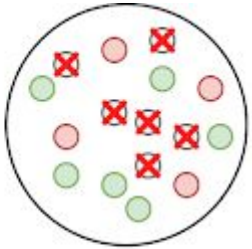
Recommender systems with risk control

Item Pool



$\mathcal{I}$

Filtering Step



$$T_\lambda(U) = \{i \in \mathcal{I} : s(U, I = i) \geq \lambda\}$$

Re-ranking Step

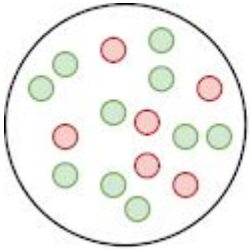


$$S_\lambda(U, k) = \{i_j \in \pi(T_\lambda(U)) : j \leq k\}$$

Risk function: $$R_H(S_\lambda(U, k)) = \frac{|\{i \in S_\lambda(U, k) : H(U, I = i) = 1\}|}{|S_\lambda(U, k)|}$$

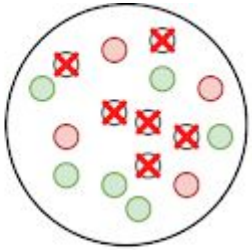
Recommender systems with risk control

Item Pool



$\mathcal{I}$

Filtering Step



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Re-ranking Step



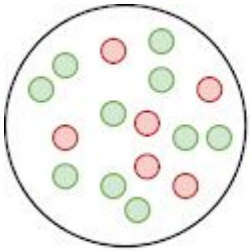
$$S_\lambda(U, k) = \{i_j \in \pi(T_\lambda(U)) : j \leq k\}$$

How can we find an **optimal threshold λ** keeping the (expected) risk under a given level α?

$$\lambda^* = \arg\max_{\lambda \in \Lambda} \mathbb{E}\left[\ell(S_\lambda(U, k))\right] \qquad \text{s.t.} \quad \mathbb{E}[R(S_\lambda(U, k))] \leq \alpha$$

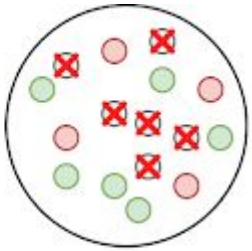
Recommender systems with conformal risk control [13]

Item Pool



$\mathcal{I}$

Filtering Step



$$T_\lambda(U) = \{i \in \mathcal{I} : s(U, I = i) \geq \lambda\}$$

Re-ranking Step



$$S_\lambda(U, k) = \{i_j \in \pi(T_\lambda(U)) : j \leq k\}$$

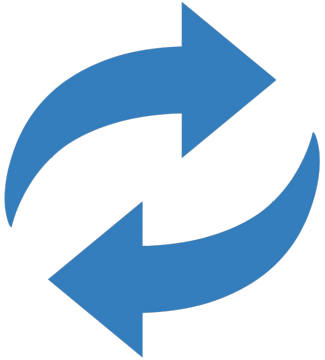
$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}(\lambda) + \frac{1}{n+1} \leq \alpha\right\}$$

For any α, we achieve:

$$\mathbb{E}[R(S_{\hat{\lambda}}(U, k))] \leq \alpha$$

(assuming R decreases monotonically with λ)

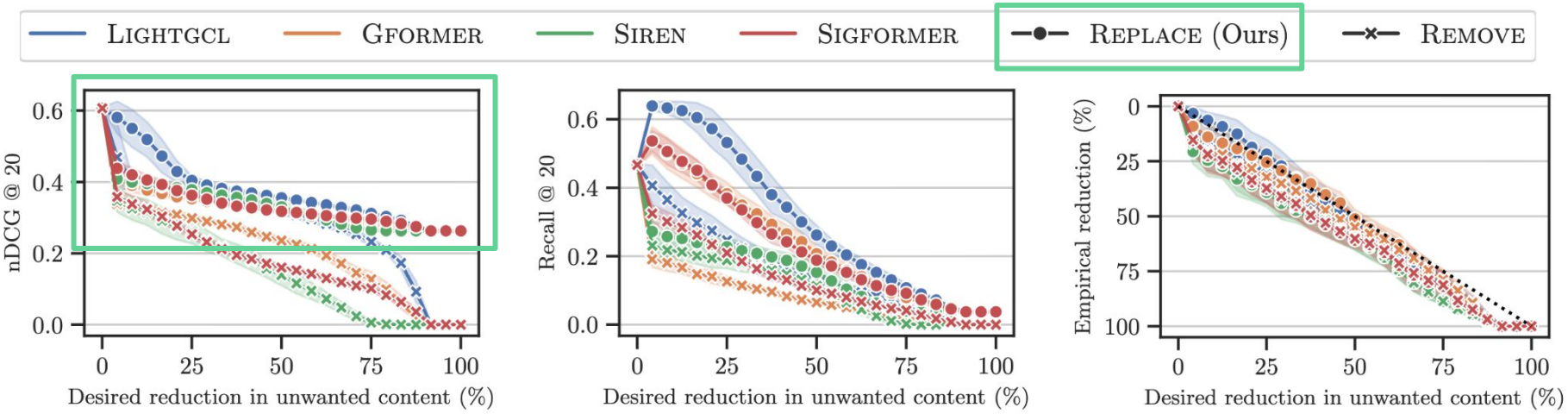
The CRC guarantees are model-agnostic and distribution-free.

Risky  Safe

Conformal risk control reduces our available items.

Safe item definition: $W_{\%}(u,i) > 50\% \ \wedge \ H_{1st}(u,i) = 0$

Main results

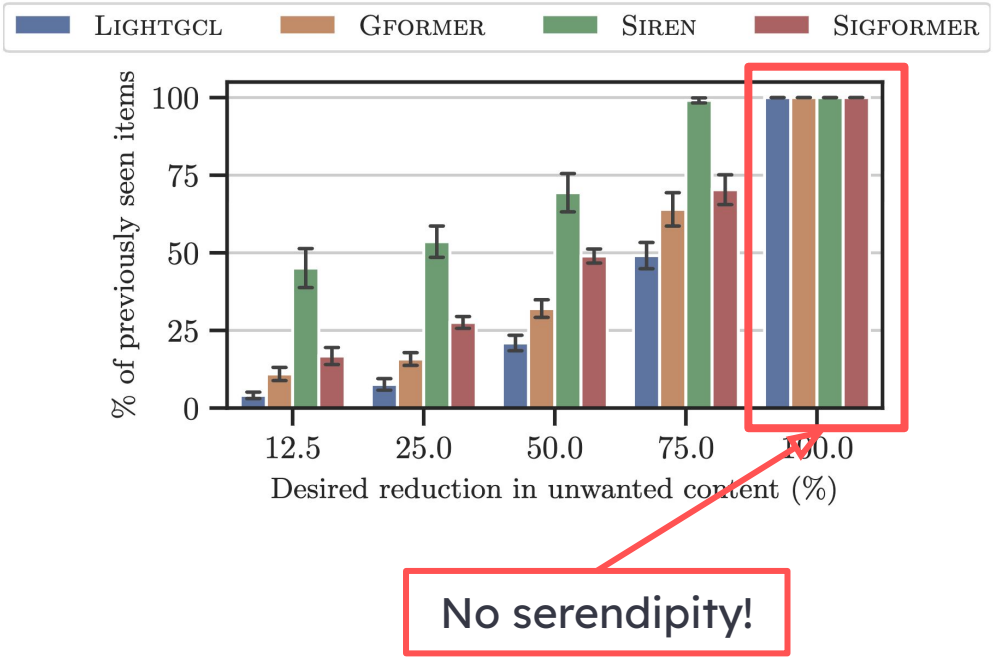


nDCG@20

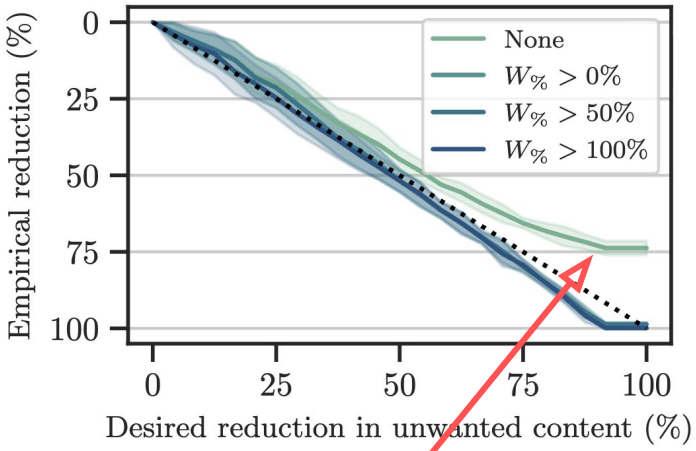
Recall@20

Expected reduction of risk

Additional insights



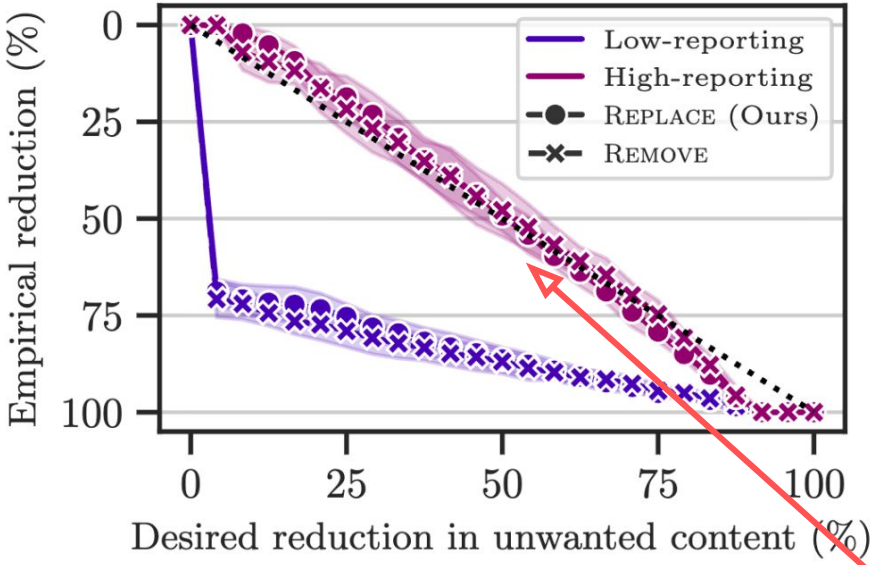
No serendipity!



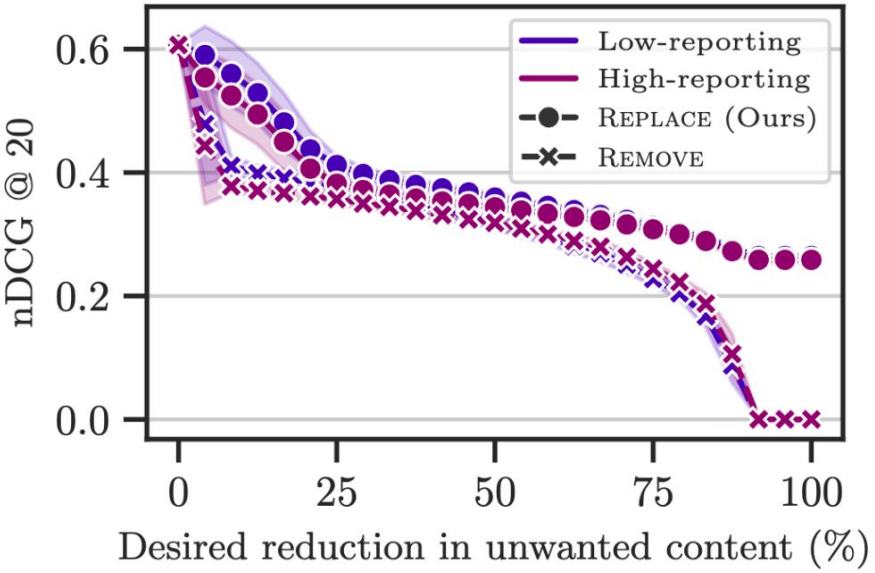
Break the risk control guarantees!

Chapter 3.

A nagging thought



Expected reduction of risk

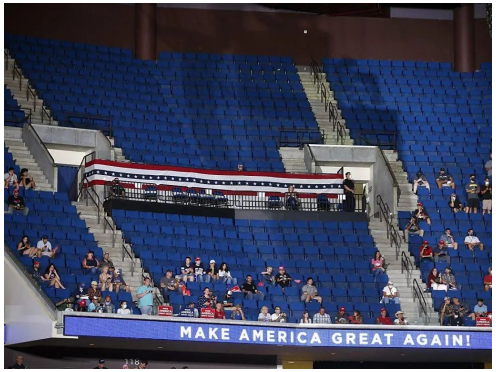


nDCG@20

High-reporting users **bias** the risk-controlling procedure

Algorithmic collective action

Groups of individuals coordinating to strategically steer a platform's machine learning algorithms to serve collective (positive or negative) goals [14,15,16].



#TulsaFlop (2020)



#DeclineNow (2019)



Mozilla Rally (2021)

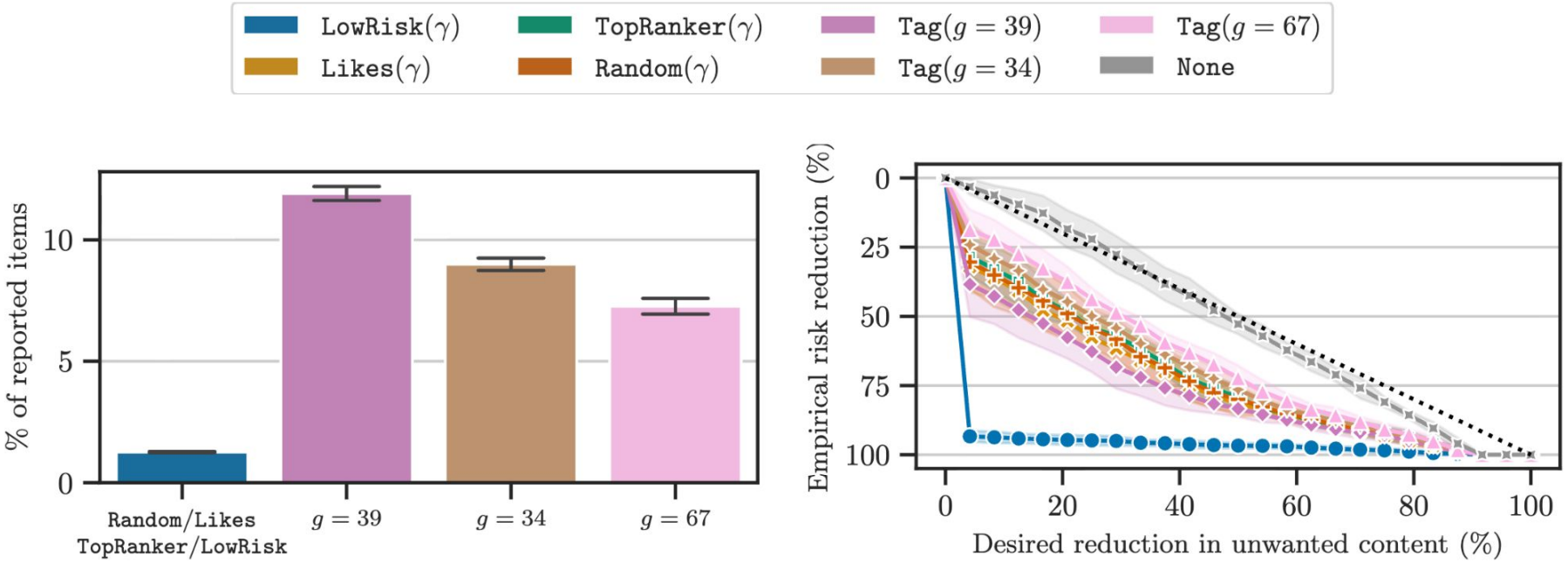


Review bombing (~2020)

What would happen if users organize to **systematically** use the “**Not Interested**” feature to **alter** the recommender **behavior**?

Chapter 4.

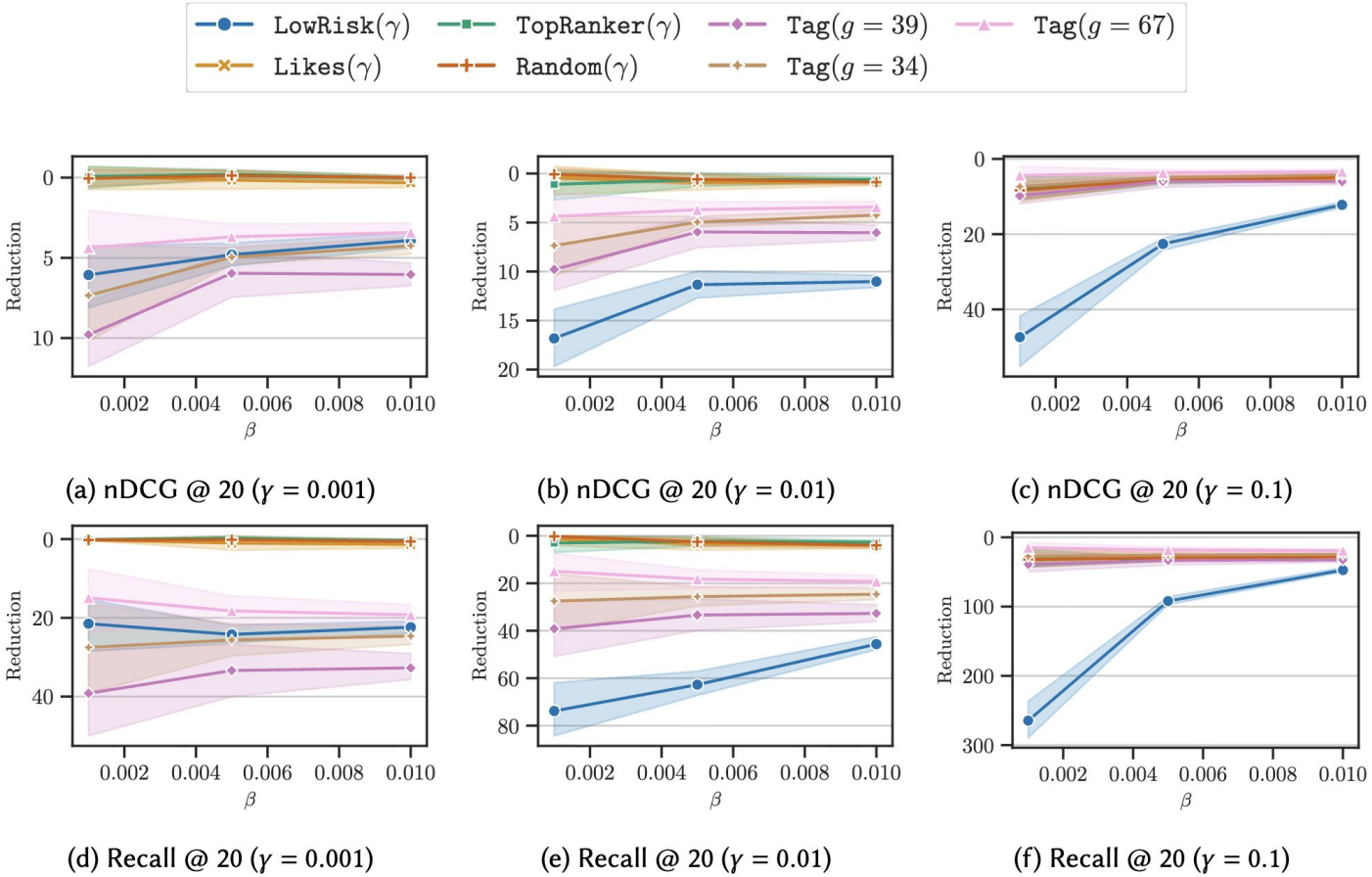
The risk analysis



(a) Frequency of reported items.

(b) Expected risk for standard users.

Effective adversarial strategies should target **low-risk items**.



(a) nDCG @ 20 ($\gamma = 0.001$)

(b) nDCG @ 20 ($\gamma = 0.01$)

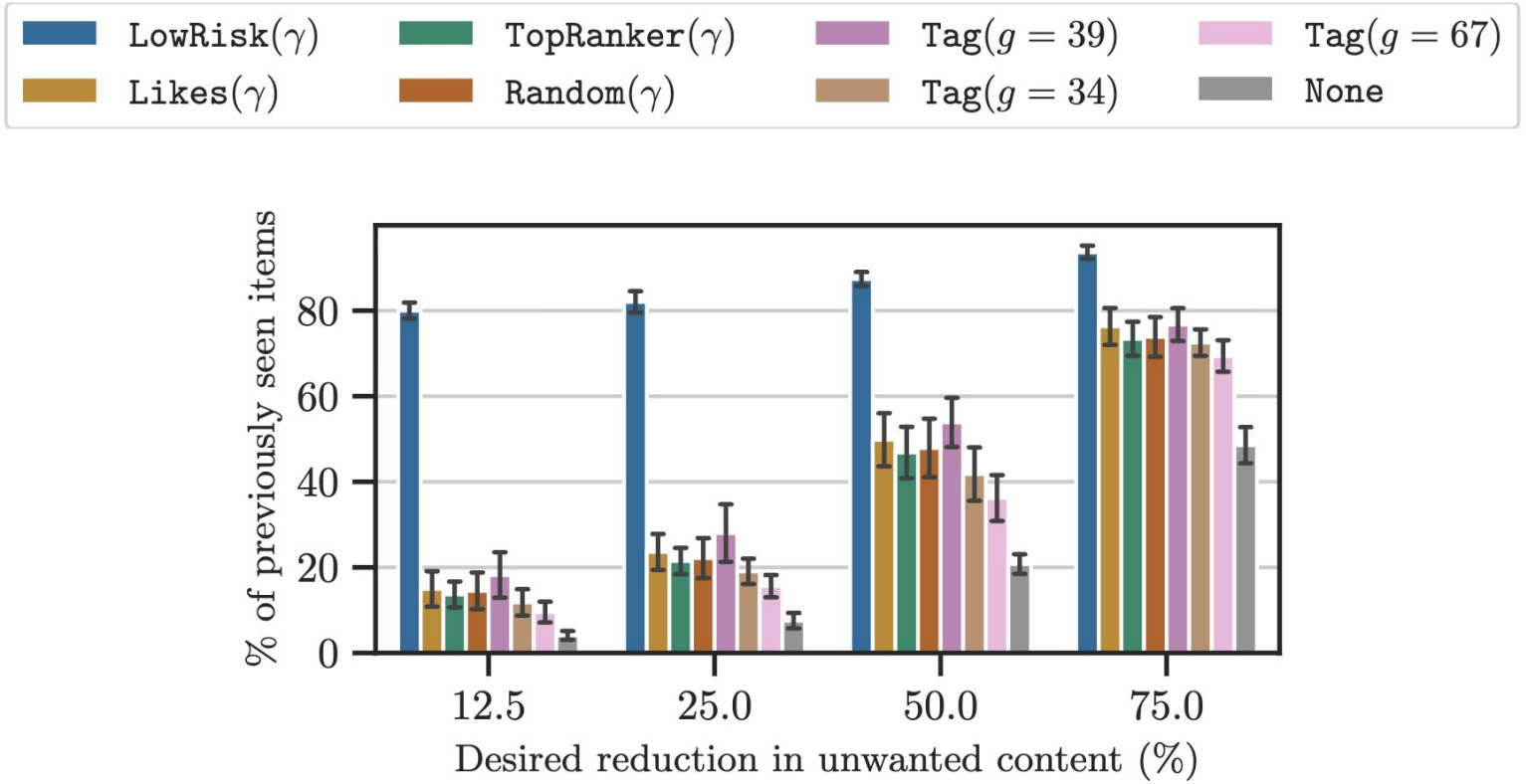
(c) nDCG @ 20 ($\gamma = 0.1$)

(d) Recall @ 20 ($\gamma = 0.001$)

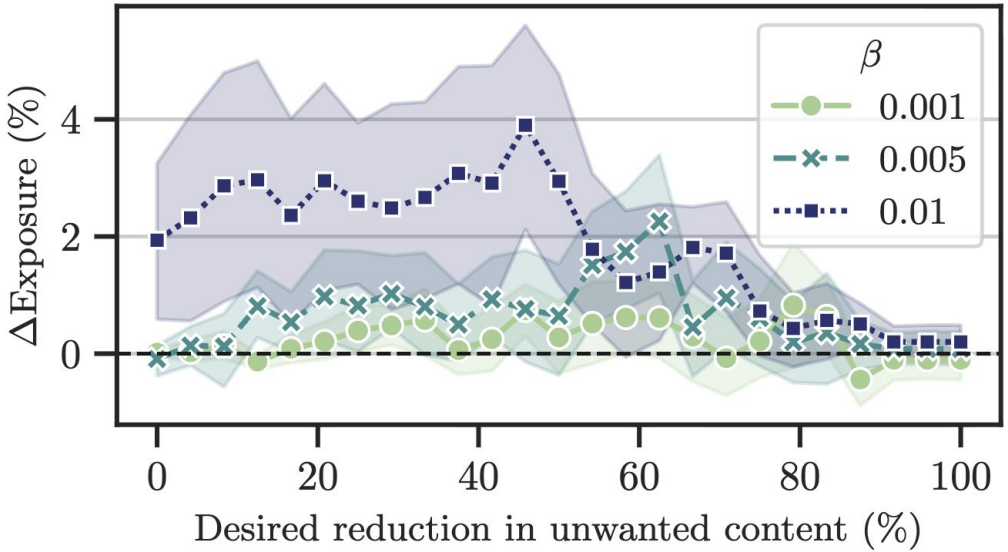
(e) Recall @ 20 ($\gamma = 0.01$)

(f) Recall @ 20 ($\gamma = 0.1$)

Few adversaries can cause a **disproportionate impact** on performance.

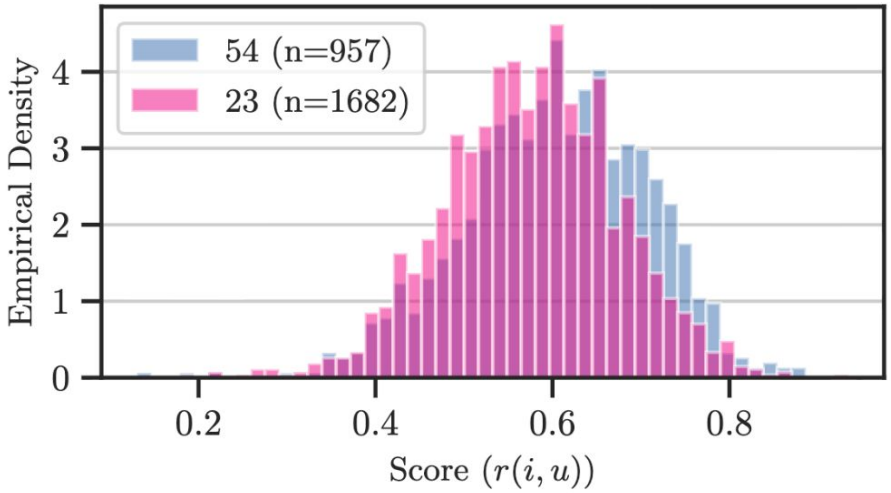


Adversarial strategies **substantially increase** repeated recommendations.

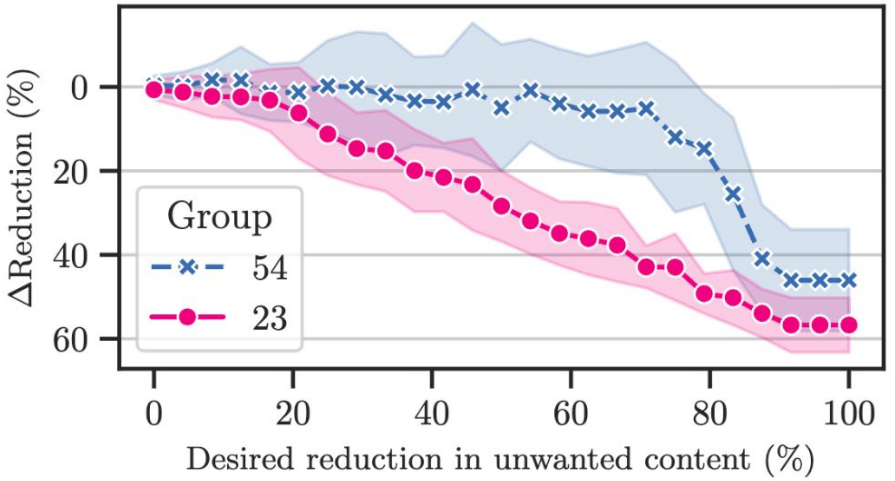


Difference in top-k exposure for group g = 34 in Kuaishou, comparing Random(γ) and Tag(g = 34).

Adversarial reporting **cannot selectively alter** group exposure.



(a) Empirical density of the risk scores



(b) Reduction in exposure in top-k

Risk control might **amplify pre-existing biases** in the learned risk model.

Chapter 5.

Our novel solution

Personalized thresholds



$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}(\lambda) + \frac{1}{n+1} \leq \alpha\right\}$$

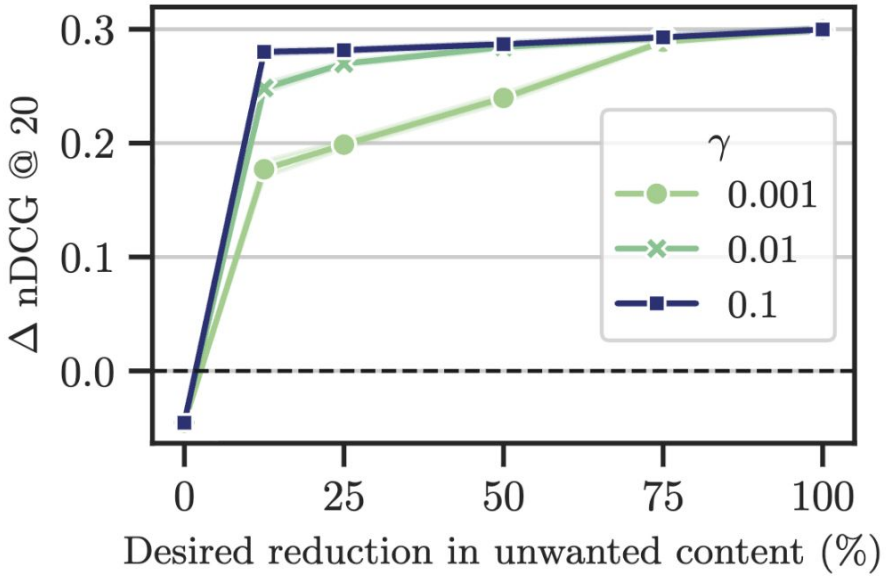
Population-level risk

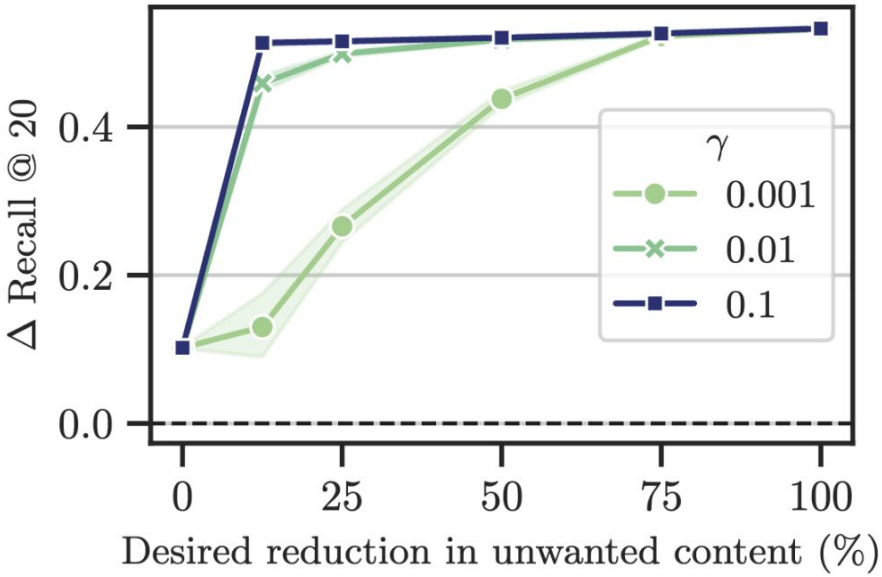


$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}(\lambda, U) + \frac{1}{n+1} \leq \alpha\right\}$$

User-level risk

Personalized thresholds mitigate the effects of adversarial collective actions





Do we end-up retreating into perfectly-safe, highly-personalised, mathematically-guaranteed **echo chambers**?

RecSys ‘25
paper

RecSys ‘25
code

FAccT ‘26
paper

FAccT ‘26
code

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References

[1] De Toni, G., Purificato, E., Gomez, E., Passerini, A., Lepri, B. and Consonni, C., 2025, September. You Don't Bring Me Flowers: Mitigating Unwanted Recommendations Through Conformal Risk Control. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems* (pp. 492-502).

[2] De Toni, G., Consonni, C., Purificato, E., Gómez, E. and Lepri, B., 2026, June. With a Little Help From My Friends: Collective Manipulation in Risk-Controlling Recommender Systems. In The 2026 ACM Conference on Fairness, Accountability, and Transparency (pp. 4712-4732)

[3] Aridor, G., Jiménez-Durán, R., Levy, R.E. and Song, L., 2024. The economics of social media. Journal of Economic Literature, 62(4), pp.1422-1474.

[4] Braghieri, L., Levy, R.E. and Makarin, A., 2022. Social media and mental health. American Economic Review, 112(11), pp.3660-3693.

[5] Sala, A., Porcaro, L. and Gómez, E., 2024. Social media use and adolescents' mental health and well-being: An umbrella review. Computers in Human Behavior Reports, 14, p.100404.

[6] Liu, A., Wu, S. and Resnick, P., 2024, May. How to train your youtube recommender to avoid unwanted videos. In Proceedings of the International AAAI Conference on Web and Social Media (Vol. 18, pp. 930-942).

[7] Tomlein, M., Pecher, B., Simko, J., Srba, I., Moro, R., Stefancova, E., Kompan, M., Hrckova, A., Podrouzek, J. and Bielikova, M., 2021, September. An audit of misinformation filter bubbles on YouTube: Bubble bursting and recent behavior changes. In Proceedings of the 15th ACM Conference on Recommender Systems (pp. 1-11).

[8] Mozilla Foundation, 2021. YouTube Regrets: A crowdsourced investigation into YouTube's recommendation algorithm. Report (pp. 1-39). https://assets.mofoprod.net/network/documents/Mozilla_YouTube_Regrets_Report.pdf.

[9] McCrosky, J., Malla, R., Tanskanen, A. and Camargo, C., 2026, April. Does This Button Work? Investigating YouTube's Ineffective User Controls. In Proceedings of the ACM Web Conference 2026 (pp. 1563-1573).

References

[10] Eslami, M., Karahalios, K., Sandvig, C., Vaccaro, K., Rickman, A., Hamilton, K. and Kirlik, A., 2016, May. First I" like" it, then I hide it: Folk Theories of Social Feeds. In Proceedings of the 2016 CHI conference on human factors in computing systems (pp. 2371-2382).

[11] Chang, S., Wang, Y., Diao, Z., Montazer, A., Zhang, J., Misra, J., Hong, D., Margolin, T., Bansod, S. and Han, N., 2026, September. Mend the Measurement Gap: Latent User Preference Modeling for Short-Form Video Recommendation. In Proceedings of the 20th ACM Conference on Recommender Systems (pp. 1291-1297).

[12] Gao, C., Li, S., Zhang, Y., Chen, J., Li, B., Lei, W., Jiang, P. and He, X., 2022, October. Kuairand: An unbiased sequential recommendation dataset with randomly exposed videos. In Proceedings of the 31st ACM international conference on information & knowledge management (pp. 3953-3957).

[13] Angelopoulos, A., Bates, S., Fisch, A., Lei, L. and Schuster, T., 2024, May. Conformal risk control. In International conference on learning representations (Vol. 2024, pp. 55198-55218).

[14] Hardt, M., Mazumdar, E., Mendler-Dünner, C. and Zrnic, T., 2023, July. Algorithmic collective action in machine learning. In International Conference on Machine Learning (pp. 12570-12586). PMLR.

[15] Sigg, D., Hardt, M. and Mendler-Dünner, C., 2025, April. Decline now: A combinatorial model for algorithmic collective action. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (pp. 1-17).

[16] Tricomi, P.P., Tarahomi, S., Cattai, C., Martini, F. and Conti, M., 2023, July. Are we all in a truman show? spotting instagram crowdturfing through self-training. In 2023 32nd International Conference on Computer Communications and Networks (ICCCN) (pp. 1-10). IEEE.